Biofilm Growth and Stress in 3D

Videos of the growth model: verification problem, seed position, growth law, tooth-like collar

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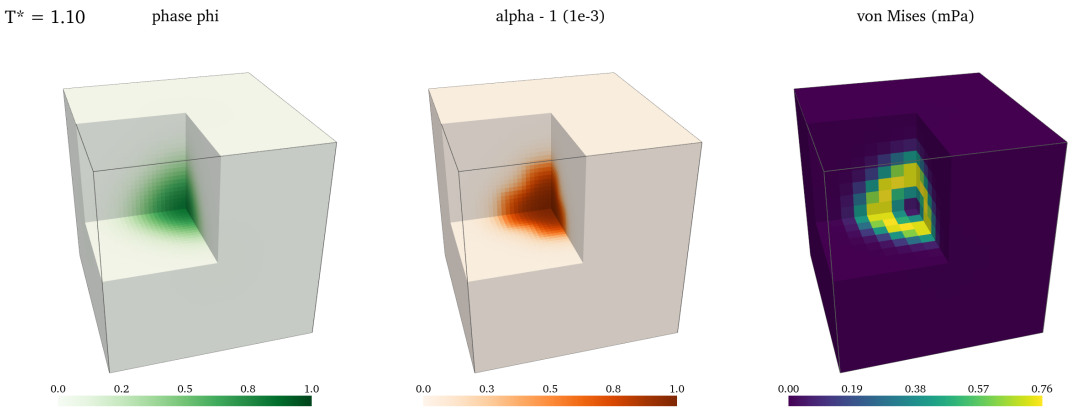
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What the videos show

- Biofilm phase ϕ, growth $\alpha - 1$ with $\mathbf{F}_g = \alpha \mathbf{I}$, and the von Mises stress, in 3D over time.
- Growth model of Klempt et al. (2024): $\dot{\phi} = \beta \nabla^2 \phi + k_\alpha \alpha$ (plus the bound on ϕ), $\dot{\alpha} = k_\alpha \phi$; stiffness $E(\phi^2 + f)$.
- Homeostatic-pressure law (added in this work): growth is scaled by $g = \min\big(1, \max(0,\, 1 - p/p_h)\big)$, $p = -\frac{1}{3} \operatorname{tr} \boldsymbol{\sigma}$.
- Four cases:
  1. the verification problem of the thesis (seed in the centre of the 2 mm cube),
  2. the same model with the seed in a corner, on an edge, or as two seeds that meet,
  3. the corner seed with and without the growth law,
  4. a tooth-like collar with synthetic four-species data (Fritsch et al. 2025).
- Only case 1 is a thesis result. Cases 2–4 are demonstrations with the same model. **No experimental data are used.**

1. Verification problem: seed in the centre of the cube

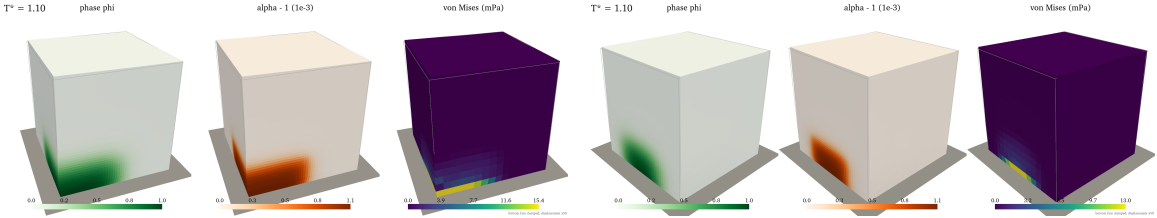


$\beta = 0.02\,\mathrm{mm}^2/T^*$, $k_\alpha = 10^{-3}/T^*$, $E = 10\,\mathrm{Pa}$, $\nu = 0.49$, $T^* = 1.1$; one octant cut away. Field on 48^3 (reference solution), stress on 16^3.

Seed von Mises at $T^* = 1.1$: 6.19×10^{-4} Pa (thesis, 16^3: 6.18×10^{-4} Pa); largest value 0.79 mPa, converged from 24^3 on. The stress forms a ring at the edge of the seed. ▶ video:

assets/anim_wp3_seed_3d.mp4

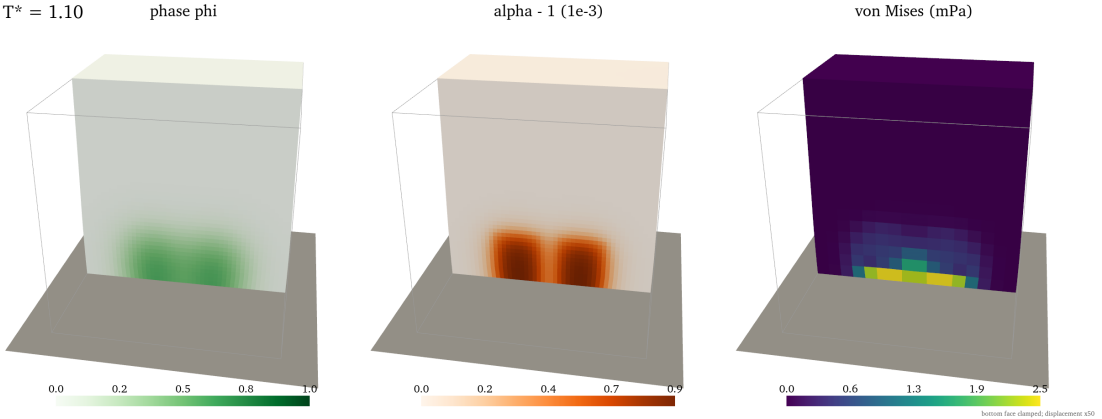
2. Seed in a corner and on an edge



Same model and seed volume as case 1; bottom face clamped (the substrate). Left: corner (three faces), right: edge (two faces). The biofilm spreads along the walls; the stress concentrates in the biofilm next to the clamped face: the mean over its bottom quarter millimetre is 7.6 mPa (corner) and 6.4 mPa (edge), 10 to 12 times the seed stress of the centre case (0.61 mPa) and about 5 times the mean of the layer above it; these means change by less than 1 % from 24^3 to 48^3. The pointwise maximum sits where the clamped face meets a free wall and grows with refinement (16 to 26 mPa, about $h^{-0.4}$), so the colour scale stops at the 95th percentile, about 15 mPa, which does not depend on the mesh.

▶ video: `assets/anim_seed_corner_3d.mp4` ▶ video: `assets/anim_seed_edge_3d.mp4`

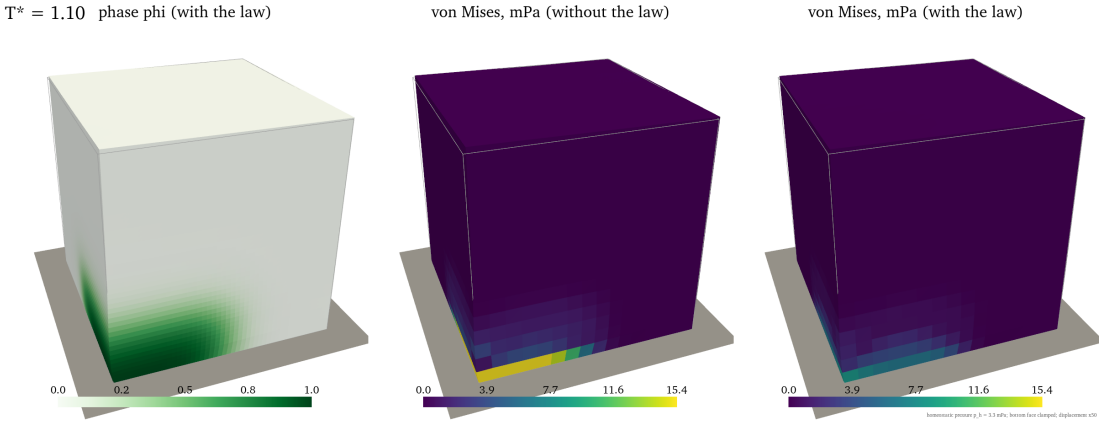
2. Two seeds that grow into each other



Two seeds of 0.5 mm on the clamped bottom face, 0.125 mm apart; front half cut away; same model as case 1. The two seeds join, and the stress is highest along the clamped face under both of them. ▶

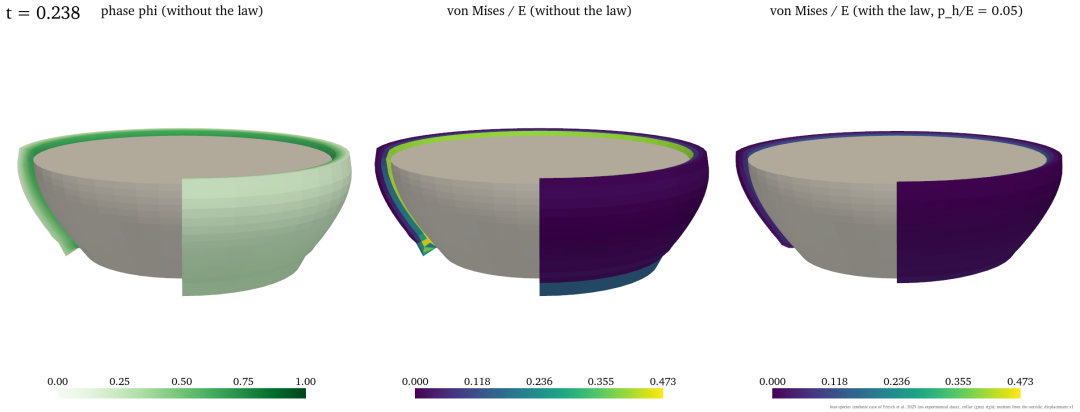
video: `assets/anim_two_3d.mp4`

3. The growth law removes the concentration



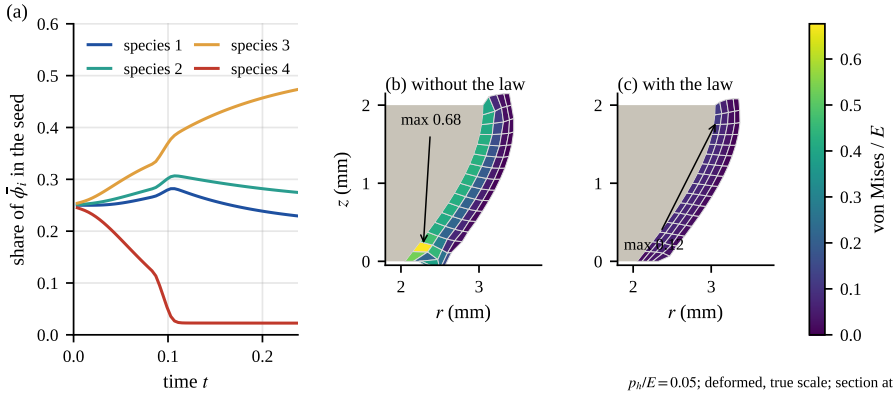
Corner seed, same colour scale in both stress panels. Without the law the stress in the biofilm next to the clamped face reaches about 15 mPa (95th percentile, the top of the colour scale); with the law it stays well below. $p_h = 3.3$ mPa is chosen by me (0.4 times the 95th percentile of the pressure without the law), not taken from a paper. ▶ video: `assets/anim_law_3d.mp4`

4. Tooth-like collar with synthetic four-species data



A 0.25 mm biofilm layer on a crown-shaped collar (radius 2.05 to 3.05 mm, 2 mm high), inner face bonded, nutrient from the outside. Point model: the four-species case of Fritsch et al. (2025) with their true parameters (synthetic, no experimental data). One quarter is cut away. Growth is large here (displacement up to 0.31 mm), drawn at true scale. ▶ video: `assets/anim_cervical_f4_3d.mp4`

4. Section and composition



(a) Species 3 takes over the seed (share 0.47 at the end), species 4 almost disappears (0.02). (b), (c) Section at 45°: without the law the von Mises stress reaches $0.68E$ at the bonded base; with $p_h/E = 0.05$ the largest value is $0.12E$.

Summary

- Case 1 reproduces the thesis value of the seed stress (0.2 % difference) and shows it in 3D.
- Moving the seed to a wall breaks the symmetry; the stress then concentrates in the biofilm next to the fixed surface, 10 to 12 times the seed stress of the centre case (mean over the bottom quarter millimetre; the pointwise maximum at the corner is mesh-dependent and is not quoted).
- The homeostatic-pressure law removes this concentration, in the cube and on the tooth-like collar (0.68 to 0.12 times E).
- Synthetic data only (Fritsch et al. 2025) keep the demonstration independent of the calibration.
- Next: a real tooth or implant surface, and the calibrated parameters of Chapter 3 at Keio.

Appendix A: notation and equations

ϕ	biofilm phase (volume fraction)
α	growth, $\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$; growth is $\alpha - 1$
β, k_α	diffusion of ϕ, growth rate (Klempt et al. 2024, Eq. 34/36)
$E(\phi^2 + f)$	stiffness, $f = 10^{-3}$ (void)
p, p_h	pressure $-\frac{1}{3}\operatorname{tr}\boldsymbol{\sigma}$, homeostatic pressure
$\phi_i, \psi_i, \bar{\phi}_i = \phi_i\psi_i$	point model (Klempt et al. 2026): volume fraction, living fraction
c^*	nutrient scale of the point model (25 in the four-species case)

Cases 1–3: $\dot{\phi} = \beta\nabla^2\phi + k_\alpha\alpha - P\,[\max(0, \phi - 1) + \min(0, \phi)]$, $\dot{\alpha} = k_\alpha\phi\, g$ ($g = 1$ without the law), $P = 5$.

Case 4: growth from the point model in the unified element of this work, scaled by the same g.

Appendix B: settings

case	geometry	mesh (field / stress)	support	source of the values
1	cube, seed in the centre	48^3 / 16^3	rigid body	thesis (Ch. 4)
2	cube, corner or edge seed	48^3 / 16^3	bottom clamped	thesis values
3	cube, corner seed	48^3 / 16^3	bottom clamped	p_h chosen by me
4	collar (quarter)	$4 \times 12 \times 16$	inner face bonded	Fritsch et al. 2025

Small strain, hex8 with B-bar in cases 1–3; finite strain in case 4. Videos: `assets/*.mp4`; scripts: `ansys_usermat/apdl/anim_*.py`, `p3/uel/anim_cervical_3d.py`.